

A descriptive analysis of significant factors influencing the behavioral intention to adopt robot kiosks/service robots in service stores: Integrating the UTAUT and higher-order SERVQUAL approach

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ABSTRACT

Robot kiosks have emerged as a promising innovation with the potential to enhance service delivery. However, their adoption remains limited, underscoring the need for a deeper understanding of the factors influencing their acceptance and use. Moreover, the country has been adopting and promoting the United Nations Sustainable Development Goals by considering these AI-enabled robots among service industries, yet no studies have yet evaluated the satisfaction, adoption, and overall service quality of these robots. This study investigated the growing adoption and impact of robot kiosks across various service sectors in the Philippines, addressing the critical need for technological advancement in enhancing service delivery. Being the first study to be conducted among robot kiosks and service robots in the Philippines, the objective was to analyze the factors influencing the acceptance, adoption, and overall service quality of robot kiosks and service robots with data collected from 412 users in service industries. Utilizing the integrated Unified Theory of Acceptance and Use of Technology (UTAUT) model and SERVQUAL 5 dimensions higher-ordered structural equation modeling approach, the study found that tangibles (β : 0.807), empathy (β : 0.782), responsiveness (β : 0.765), assurance (β : 0.711), reliability (β : 0.646), and behavioral intention (β : 0.104) significantly affected customer satisfaction. Moreover, performance (β : 0.162) and effort expectancy (β : 0.156), social influence (β : 0.112), and service quality (β : 0.549) are key determinants of adoption. However, facilitating conditions (p -value = 0.283) were not found to be a significant predictor. The research highlights that well-designed, responsive, and empathetic robot kiosks improve customer engagement and satisfaction, but challenges like limited infrastructure may hinder broader implementation. These findings suggest that strategic deployment of robot kiosks can drive technological innovation, enhance service quality, and support socio-economic development in developing countries like the Philippines. The study's significance lies in providing actionable insights for businesses and policymakers to optimize robot kiosk usage and advance the integration of AI technologies in the service sectors.

1. Introduction

The recent development of technology encourages many entrepreneurs to get creative and find different ways to engage with customers. One of the commonly used technologies, especially in the service sector is robot kiosk [1]. The use of robot kiosks and service robots (Fig. 1) are growing in acceptance in several industries, including retail, hotels, and healthcare, among other service industries [2]. In retail industries, stores frequently use robot kiosks to help with self-checkout, product

recommendations, and consumer concerns, leading to significant changes in how the service environment is structured and experienced [3]; while service robots provide deliveries within the vicinity or among hotel room deliveries [2].

According to Song and Kim [4], robot kiosks offer several benefits in retail and service industries. They save labor costs, enhance productivity by facilitating better collaboration between humans and artificial intelligence (AI), create enjoyable shopping experiences, and automate inventory monitoring systems. Robot kiosks/service robots stand out

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Fig. 1. A delivery service robot operating in a supermarket.

from other computer technologies due to their unique capacity to simulate human emotions, engage in complex interpersonal interactions, and utilize facial recognition technology to detect and respond to users' emotions [5]. Nonetheless, Bongomin et al. [6] explained how customer acceptance and utilization of which provides the barrier in service industries. This means that there needs to be an exploration on the adoption, especially overall service qualities since these robots have been currently deployed for service encounters.

In the hospitality industry, hotel and restaurant service robots are frequently used to facilitate check-in procedures, offer room service, and communicate with guests. Jang and Lee [2] anticipated that service robots will play a critical role in ensuring the restaurant industry's long-term viability, particularly given the broad acceptance of contactless services brought on by the COVID-19 breakout. Tuomi et al. [7] added that they have continuously evolved to provide more effective human-like services and are now receiving significant attention from service firms in the tourism and hospitality sectors. On the other hand, robot kiosks among healthcare industries may assist with patient check-ins, appointment scheduling, and providing medical information. Holland et al. [8] indicated that the health industry, which has traditionally received less attention, is now seeing new opportunities for the development of service robots. The researchers indicated that the COVID-19 pandemic accelerated the advancement of service robots in the healthcare sector.

Moreover, the necessity for technological developments to fulfill evolving consumer demands in the digital era has driven recent trends and adjustments in robot kiosks across numerous industries [9]. In the study of Kunz et al. [10], the researchers found that humanoid robots

have extraordinary capacities, as demonstrated by Pepper, a well-known humanoid robot in hotel settings. They display greater independence and adaptability by completing tasks on their own while continuing to establish service expectations. The service robots utilized their ability to collect and use existing knowledge, which not only improved operational efficiency but also promoted service innovation. Furthermore, Gonzalez-Aguirre et al. [11] emphasized the development of robot kiosks in the United States, with sales volume expected to reach USD 27 billion in previous years, suggesting a growing need for service robots in various industries. The study stressed the importance of maintaining and advancing service robots, such as robot kiosks as they are expected to become increasingly essential in daily life in the future.

Other studies considered the assessment of progress among manufacturing firms due to automation with robots. Duarte et al. [12] implicated that human-robot interaction is still needed since the full automation may not be as efficient. Therefore, there needs further assessment on the users' perspective and adoption to understand what areas need improvement. Sanchez et al. [13] further supported how artificial intelligence, which runs the service robots and robot kiosks operate, have high sustainability on the linear projects. Moreover, it was added that the automation of these types of projects are indispensable for sustainability. Recent advances in service robot technology throughout the world have sparked the Philippines' interest in producing similar innovations on a local scale. As automation becomes increasingly significant, the Philippines is considering deploying service robots to enhance various areas such as hotels and hospitals [14]. Currently, it is projected that service robots' market would reach \$80.6 million USD with volume of \$55 million USD until about \$208 million USD by 2029 [15].

As automation gains traction in the Philippines, a notable shift towards integrating robotic solutions across various industries exists. The deployment from SM (a popular mall in the Philippines) of robot concierges, called SAM (Slightly Autonomous Machine), across its malls represents a significant advancement in the country's adoption of robot kiosks [16]. This initiative underscores the increasing focus on leveraging technology to enhance customer service and engagement within the retail sector. By pioneering the deployment of robot concierges and embracing automation, SM sets a precedent for other businesses to follow, driving further advancements in robot kiosk technology and reshaping the retail landscape nationally [17]. Their counterpart, Robinsons' Place have also deployed robot kiosks as evidenced among supermarkets (Fig. 1). This is because the current operating hours of the mall is from 10am, sometimes 9am until 10pm. With lunch breaks and in-between work breaks, there would be times where there would not be any employees for service. In accordance, these employees would need to be compensated for overtime. Therefore, the addition of these robot kiosks and service robots would reduce the need for concierge services as most often, customers ask for direction, locations, and availability, which the robots could accommodate to complete the mall experience [16].

Another significant introduction is Jollibee's automated meal-serving robots, known as Jollibots. As Candice Urot [18] highlighted, Jollibee aims to facilitate customer transactions, prioritizing efficiency, and convenience. The launch of Jollibots reflects a broader trend of incorporating robotic solutions into retail and service industries, illustrating the Philippines increasing adoption of automation to enhance efficiency and customer satisfaction. Ang et al. [19] provided evidence how there is an increasing rate of robot availability in the Philippines, which is correlated with the popularity of service robots in the travel, retail, and hospitality industries. This growing adoption of robots reflects a broader trend driven by the desire for minimal or no human errors in automated systems, which are commonly associated with technological advancements. The study noted that clients tend to prefer quick, efficient, and innovative products and services which the robots could easily accommodate, presenting higher service quality for the transactions. However, it was also observed that people are often never

fully satisfied, as they continually desire the latest and most up-to-date offerings available.

Furthermore, the Philippines' use of service robots has shown a lot of potential to raise the standard of goods and services provided in a range of industries. Magsumbol et al. [14] indicated that AI robots may guarantee consistency and accuracy in production processes in industrial industries, resulting in higher-quality goods. These robotic systems help to improve the country's manufacturing operations' overall efficiency and dependability by automating repetitive processes and reducing mistakes. This was also evident in the study provided by Sanchez et al. [13] and Duarte et al. [12].

With several advancements made, service robots are increasingly being used in retail and other service industry settings to address a variety of business challenges. According to Ivanov and Webster [20], one of the primary reasons of implementing service robot is because of the expenses associated with hiring and retaining human staff members among service industries. The study suggested that by introducing robot kiosks and service robots, businesses may be able to reduce their dependency on human labor while also improving many elements of their operations, such as competitiveness, service quality, and operational efficiency. Moreover, a study conducted by Brengman et al. [21] examined the difference of a tablet service kiosk (TSKs) and humanoid service robots (HSRs) in retail environments. The result indicated HSRs' significant advantages over TSKs, notably in attracting the people and turning them into customers. The presence of HSRs improved the number of people looking at and stopping by the store, resulting in more unique transactions and greater spending amounts on days when HSRs were deployed. It demonstrates the growing usage of service robots in retail to increase sales and customer engagement as well as maintain competitiveness and customer satisfaction.

Rogelio et al. [22] indicated that the Philippines has limited resources for service robots due to a variety of regulatory challenges, flaws in the country's technology infrastructure, and economic limitations. With very little funding set for robotics research and development, the nation may find it difficult to keep up with breakthroughs made worldwide in the industry. In addition, the Fourth Industrial Revolution (FIR) represents a shift towards automation and data exchange in manufacturing technologies. As Dadios et al. [23] highlighted, both the Philippines and the global landscape are poised to experience significant growth opportunities in productivity and the development of new business models, goods, and services. Although these technologies are currently utilized in the Philippines, improvements are still needed, particularly in robot services, to maximize the potential of their adoption. Kim et al. [24] expressed that the Philippines' limited access to robotics and distinct economic circumstances mean that the country will not experience the FIR in a comparable way. The FIR may be perceived as an actual risk to the Philippines since it can potentially replace human labor with fewer service robots. The study suggested that the Philippines must take another approach while dealing with the FIR and consider its economic situation. Moreover, Dadios et al. [23] added that the Philippines should embrace foreign investment and trade to facilitate access to new AI robots and promote competitiveness, especially in information and communication technology (ICT) to effectively adapt to the rapidly changing technical landscape. With this, several gaps are needed to be assessed on a holistic point of view to undermine whether robot kiosks and service robots provide satisfaction on the services and how adoptable are people with it. The research gap, problems, and this study novelty are listed as:

- The current infrastructure in the Philippines may be limited but continuous deployment are present among service and retail industries. Yet, the service quality, satisfaction, and acceptance of people in the country has yet been assessed.
- The current facilitation of automations is on the rise in the service and retail market, but no studies in the Philippines have yet assessed

the overall perception of people on the service that the robots provide.

- The Philippines is a country trying to adhere with the United Nation Sustainable Development Goals, considering the development to be a smart city by utilizing and promoting the use of AI-enabled robots. Nonetheless, the rise of these technologies has yet been thoroughly considered on the Philippine users' perspective.

This study has therefore considered this research gaps and problems as inspiration to assess the overall adoption, satisfaction, and service quality of the robot kiosks and service robots. Being the first study covering this in the country, a holistic assessment was made based on frameworks to ground the analysis. Various studies have examined the use of service robots by employing the Unified Theory of Acceptance and Use of Technology (UTAUT) framework alongside Service Quality measures. The study of Meyer-Waarden et al. [25] utilized a SERVQUAL framework to identify the key factors influencing acceptance and intention to reuse of chatbots. The limitation of this study focuses on the need for alternative measurements targeted to service settings. They presented that SERVQUAL 5 dimensions alone cannot holistically characterize technology adoption. The study suggested looking into the changing nature of human-robot interactions in service contexts as well as how user perceptions and demands could shift over time. Moreover, Yeoh and Chin [26] employed UTAUT model to evaluate the variables affecting Malaysians' decision to use home healthcare robots (HHRs). The study's findings may not be broadly applicable due to several limitations. The sample size and the focus on respondents only among those providing at-home care for elderly or handicapped individuals restricted the generalizability of the results. Additionally, the study's exclusive focus on Malaysia limits its ability to consider larger socioeconomic or cultural factors influencing HHR adoption. That is, developing countries like the Philippines have already started adopting the service robots, but has yet been assessed for acceptance, satisfaction, and service quality. This could help provide comparative output. Additionally, Pande and Gupta [27] utilized the UTUAT model to evaluate and validate a theoretical framework for assessing the acceptability of service robot usage (SRUA) within the context of Indian restaurant services. However, the study's narrow focus on Indian restaurant settings suggested that the findings may not be universally applicable to other cultural or contextual environments since other service industries were not assessed.

In the Philippines, the use of robotic services in the hotel and food industries is still a developing field. Luo et al. [28] stated that there is a lack of thorough knowledge regarding the relationship between robotic services and customer satisfaction, despite knowing that researchers and hotel managers are paying more attention to integrating robots and AI into hotel operations. They tried to address these gaps by assessing customers' views in online evaluations of prominent hotels that use robotic services. Moreover, Rosales et al. [29] indicated that AI technology represents an exciting development in the Philippine corporate setting, but there is a huge scarcity of experienced individuals to promote its adoption. Any AI project's success is put at risk if there is insufficient talent among data engineers, data scientists, and AI engineers. As a result, education and training activities are critical to raising understanding and capability in these areas. Overall, the Philippines is still catching up on the deployment of AI systems – and highlights the need to explore on service quality, technology adoption, and satisfaction among consumers in the current market to understand what needs to be modified on the practice and technology itself.

This study aimed to analyze the variables impacting the acceptance and use of robot kiosks in the Philippines, a country that is currently managing its integration of AI and robotic technology. Utilizing the integrated UTAUT and SERVQUAL 5 dimensions as framework, higher-ordered structural equation modeling approach was employed for the holistic assessment of the causal relationship between latent variables affecting satisfaction, behavioral intention, and technology adoption. Understanding these behavioral goals is critical for organizations,

legislators, and researchers hoping to improve service quality, efficiency, and customer satisfaction by strategically deploying robotic solutions. The significance of this study lies in its potential to provide actionable insights for businesses and policymakers, informing decision-making processes related to the adoption, implementation, and improvement of robot kiosks across service sectors in the Philippines. Additionally, this research contributes to the broader literature on technology acceptance and service quality by examining the unique context of robot kiosks and their impact on customer experiences in a developing technological landscape. Both practical and managerial, as well as theoretical implications have been provided.

This paper has been arranged as (1) the introduction, providing insights on the problems needed to be addressed, research gap, novelty, research objectives, and research contribution. This is followed by (2) literature review, conceptual framework, and hypothesis build up, (3) methodology, (4) results and validity measures, (5) discussion and study implications, and (6) conclusion.

2. Literature review and conceptual framework

The adoption of robot kiosks and service robots have seen a significant rise due to technological advancements and increased accessibility. In this context, the UTAUT model could play a crucial role in measuring the acceptance and satisfaction levels of these kiosks, incorporating service quality variables. The UTAUT model, with its focus on factors like performance expectancy, effort expectancy, and facility condition, provides a comprehensive understanding of users' intentions and actual usage behavior among technology [30]. Moreover, Forgas-Coll et al. [31] have expanded the UTAUT to encompass the acceptance of new technologies. Their study not only highlighted the importance of specific factors in influencing behavioral intentions, but also underscored the practical implications of this expansion. Likewise, the study mentioned that this expansion of UTAUT emphasizes how crucial it is to consider these factors when figuring out how people accept and utilize new technologies.

Similarly, Gutsche and Schlögel [32] also introduced an expanded framework that contains service quality. Given that more technological possibilities for providing such information are becoming available as digitalization advances, the study emphasizes how these factors

connect, particularly in industries heavily reliant on new technologies. By examining these models within the framework, they were able to elucidate how technological advancements impact service quality and how this understanding can guide improvements in service delivery.

Adopting the higher-order reflective construct of service quality domains from the study of German et al. [33], the model in this study generated a total of 8 hypotheses. As suggested by Sarstedt et al. [34], a higher-order reflective construct promotes the accuracy and relevance of variables under a theory as its domain because it reduces the path analysis the multivariate tool needs to consider. To which, the integrated UTAUT and SERVQUAL 5 dimensions by Parasuraman and Berry [35] and modified by Zeithaml et al. [36] is presented in Fig. 2.

Na et al. [37] have significantly contributed to our understanding of technology adoption. They have identified performance expectancy as the degree to which an individual believes using a system will enhance their efficiency at work. This concept has emerged as a critical factor in understanding the adoption of kiosks. Moreover, Roy et al. [38] that focused on service robots in hospitals asserted that performance expectancy showed a great impact on behavioral intention. Their findings revealed a significant positive relationship between performance expectancy and customers' positive emotions. They discovered that customers' positive emotions were associated with greater acceptance of AI and reduced objections to its use during service encounters. Song and Kim [4] likewise found a significant correlation between behavioral intention and performance expectancy. Their research emphasized on the use of humanoid robots in retail settings, highlighting the increasing usage of these machines in the retail industry due to its performance. Thus, it was hypothesized that:

H1. Performance expectancy has a significant influence on behavioral intention in robot kiosk satisfaction and acceptance.

Seo [39] indicated that effort expectancy refers to a person's perception of how easy it will be to use the system. In a study by Alaiad and Zhou [40] on patients' behavioral intention toward healthcare robots, it was found that effort expectancy plays a significant role. Their results indicated that patients' perceptions of how easy it is to use healthcare robots significantly influence their behavioral intention. Kelly et al. [41] also found that effort expectancy strongly and positively impacted the behavioral intention, willingness, and adoption of AI

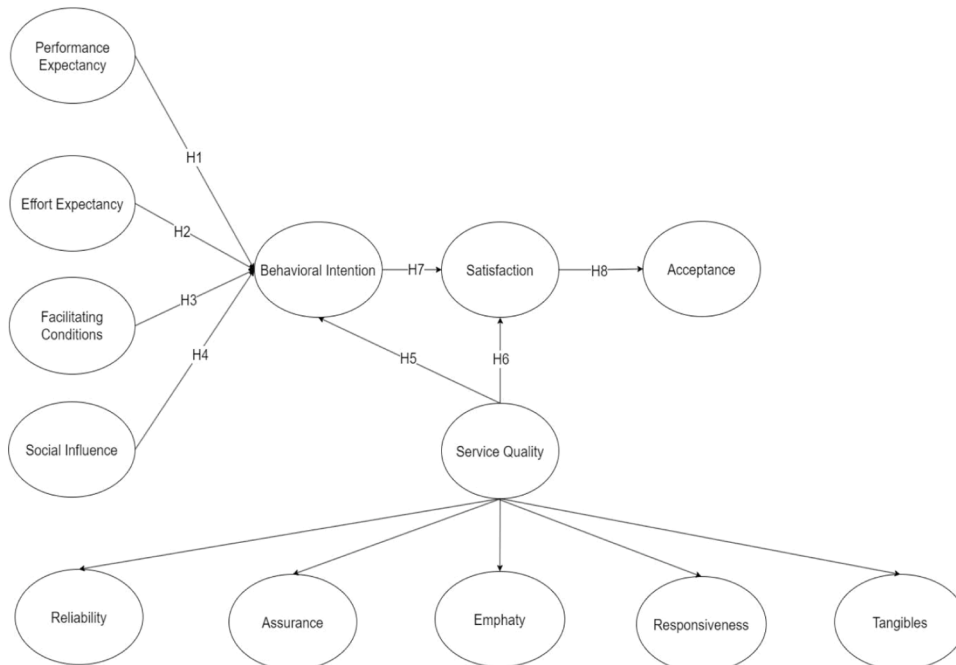


Fig. 2. Conceptual framework.

robots across different sectors. According to the study, effort expectancy refers to the level of convenience associated with using AI robots, similar to the Technology Acceptance Model construct of perceived ease of use. The utilization of this technology has the potential to enhance various aspects of human life, particularly through advancements in healthcare technology. In another study on the acceptance of AI robots in hospitality services, Roy et al. [38] demonstrated how effort expectancy influences customers' behavior. Thus, it was hypothesized that:

H2. Effort expectancy has a significant influence on behavioral intention in robot kiosk satisfaction and acceptance.

In addition to technological aspects, facilitating conditions have a considerable impact on user adoption and satisfaction with robot kiosks. El-Said and Tall [42] found that facilitating conditions increase customers' willingness to use new technologies in various contexts. Their research focused on self-service robots in the food industry, which leverage modern technology to meet customer expectations. Similarly, Seo [39] showed the significant effect of facilitating conditions on behavioral intention. The study defined facilitating conditions as individuals' belief in the availability of both organizational and technical support systems to facilitate the use of the system. It was implicated that restaurants should consistently advocate for increased investment in technology development, as found that the development of an infrastructure for technology use is an essential factor. Furthermore, Alaiad and Zhou [40] also asserted the significant influence of facilitating conditions on behavioral intentions. Their study mentioned that potential use barriers could be addressed or avoided by providing user training and help during challenges. As a result, it is established that patients' perceptions of facilitating resources, which include technical skills, workplace, time, and financial support, will impact their decision to use healthcare robotics services. Therefore, it was hypothesized that:

H3. Facilitating condition has a significant influence on behavioral intention in robot kiosk satisfaction and acceptance.

The framework also incorporated social influence as a variable in the study, aiming to offer insights into how social factors affect individuals' behavioral intention toward the acceptance and satisfaction of robot technology [43]. Alaiad and Zhou [40] highlighted the positive impact of social influence on users' behavioral intentions. Their study revealed that among all the factors measured, social influence emerged as the strongest predictor for the intention to use healthcare service robots. Given the focus on healthcare service robots, it can be inferred that the decisions of patients regarding these robots may be influenced by the opinions and recommendations of their relatives and friends. Furthermore, social influence refers to the degree to which individuals around the users adopt a new system, thereby influencing their own behavior [37]. In the same study, the acceptance of self-serving kiosks found that social influence directly correlates with behavioral intention. Specifically, the study indicated that higher levels of social influence led to increased behavioral intention to utilize menu items through robot kiosks among customers, aside from healthcare service robots, robot kiosks in the food industry have demonstrated a positive impact on behavioral intention [42]. The research further confirmed that customers' intentions to adopt new technologies are influenced by their social environment, including the opinions of friends and family. Thus, it was hypothesized that:

H4. Social influence has a significant influence on behavioral intention in robot kiosk satisfaction and acceptance.

Parasuraman and Berry [35] defined service quality as the difference between a client's expectations of a service provider and their assessment of the actual service they receive. Boulding et al. [44] examined how different aspects of a service shape people's behavioral perceptions of its quality, which impacts their future behaviors. They found that people's expectations before and during a service can affect how they perceive its quality, and that overall quality perception can influence

their behavior and intention. The results showed that when customers feel the service quality is good, it tends to positively influence their behavioral intentions. This is still evident in the current generation, which is why its relevance as a framework is still considered today [36]. In addition, Zhu [45] explored on the relationship between several robot restaurant-related aspects and customer intentions to come and return including food quality, service quality, and the impression of a high-tech environment. The study suggested that customers value the level of service that robot restaurants offer and are more likely to come back or have a good experience if they think the level of service is high. More specifically, the results presented that consumers' intentions to use service robots and return to robot restaurants are positively impacted by their assessment of the quality of the services provided.

Robot kiosks are becoming increasingly common in certain industries [46]. For example, the study of Duarte et al. [12] explained that the human-robot interaction in the manufacturing industry needs personnel conversely and full automation would not be possible since it would not be efficient. Thus, there needs to be human-robot interaction that needs evaluation for fit. Related technologies have also been implied to help increase efficiency, operation, and quality of life among users [13]. It was elaborated that these technological advancements have changed consumer interactions with businesses, leading to positive outcomes such as satisfaction, commitment, and intentions to continue using a service [47]. Research findings indicated that quality-of-service technology directly influences customer satisfaction, loyalty, and intentions, both positively and significantly [48]. Moreover, Li and Wang [49] explained that service robots play a crucial role among service industries as they provide customized services among people, providing the current trend in the market, and overall perception of service affected intention to use – especially when positive attitude among users are seen. Thus, it was hypothesized that:

H5. SERVQUAL and the dimensions have a significant influence on behavioral intention.

With significant technological advancements, particularly in AI, the service and retail industries began to use robots to perform a variety of tasks in the current generation [50]. Previous research has utilized service quality as a tool to evaluate user preferences in fulfilling their needs across multiple sectors. In this context, Morita et al. [51] employed a similar model, SERVQUAL to assess customer satisfaction with service robots in cafes. Their findings confirmed that the service quality domains provided by service robots in cafes significantly impacts customer satisfaction and their intention to return. Moreover, Shah et al. [52] examined customer satisfaction with service robots. They discovered consumers feel more engaged and are in turn more likely to accept and feel comfortable with robots when they perceive the machines are capable of automation, customization, efficiency, and precision. The relationship between perceived service quality and engagement may be strengthened when customers see robots as more human-like. Therefore, customers are more likely to accept and feel comfortable adopting AI service robots in restaurants when they are more involved with them.

In addition, a study conducted in Korea examined the adoption of personal service robots in households [53]. The research aimed to understand how people in Korea perceive and utilize personal service robots in their homes, focusing on aspects such as service quality and technology adoption using Service Quality Model and Technology Acceptance Model. The study emphasized that the system's overall quality, including its functionality and performance, significantly influenced the usefulness of the robots in their homes. This highlights the significant impact of service quality on customer satisfaction. Thus, it was hypothesized that:

H6. SERVQUAL and the dimensions have a significant influence on customer satisfaction.

Besides considering physical factors, the UTAUT model also takes into account behavioral intention when evaluating consumer

satisfaction, especially with the increasing presence of technology in retail settings. For instance, Song and Kim [4] defined robot kiosks and service robots as AI-driven humanoid machines providing personalized shopping assistance. It was indicated that customers respond more positively to these robots when they are helpful, sociable, and visually appealing, reflecting an increasing demand for better service facilitated by robots [4]. However, consumers' beliefs about interacting with robots can influence their perceptions. Therefore, creating robots with attributes like affection, assistance, and visual appeal can enhance consumer acceptance.

Similarly, Kim and Lee [53] found that consumers are more likely to continue using a personal robot if they find it useful and satisfying. Among Korean households, focused on system effectiveness, service level, and ease of technology adoption, it was seen that behavioral intentions played a significant role in customer satisfaction. Their findings implicated that consumers' intentions influence their emotional responses, which subsequently impact their acceptance of service robots, particularly in the hospitality industry [54]. Studies that have been mentioned have continually shown that consumers are more likely to employ an AI robot service in the future when they feel satisfied with it and have the intention to use it. This also holds relevant for personal robots; people are more willing to keep making use of the robot in their homes if they find it useful and are satisfied with it. Thus, it was hypothesized that:

H7. Behavioral intention has a significant influence on customer satisfaction.

There is also a significant effect between the satisfaction of consumers on service robots and robot kiosks. This is because customer satisfaction is pivotal in determining customers' adoption of a particular robot technology [55]. However, studies like that from Li and Wong [49] implicated that acceptance of these robots are still a major and crucial barrier for full deployment. This is because overall satisfaction on utility should be achieved first before customer acceptance would be positive. Since the country has yet fully adopted to these automated robots, there needs further evaluation of the current users for improvement, if any, for holistic deployment to be positive.

According to Wu et al. [56], satisfaction significantly influences the acceptance of technology. Customers may have reasons for both accepting and rejecting the automated robot service, which can influence their overall satisfaction and acceptance. This underscores the complexities of consumer decision-making and highlights the importance for different businesses to acknowledge and address both supportive and opposing factors to enhance customer acceptance and satisfaction. Similarly, Xie and Kim [57] delved into the acceptance of hotel service robots, aiming to understand the factors that make these robotic devices more favorable to users. Specific service elements consistently desired by customers was observed needed to be considered. To which, user acceptance can significantly increase as the service factor aligns with these demands.

Ding et al. [58] elaborated that for adoption to be fully accepted on these robots, there should be positive satisfaction on the human-robot interaction in terms of functionality, shorter processing time and interaction with customer, social-emotional factors are present, and that service quality is overall positive. Furthermore, Kao and Huang [59] studied how consumers responded to robots in various kinds of service instances, including full-service restaurants and limited-service institutions. They observed that when customers believed they had a satisfying connection with a robot, they were more likely to perceive it as helpful and keep a favorable attitude towards using it – in particular conditions where more assistance was offered.

Among restaurants, these robots are said to be accepted when personalization, precision, and overall service quality efficiency is achieved [52]. The overall robot engagement relates to the users' satisfaction and acceptance. Essentially, when consumers are satisfied with the purpose of robots in their service experiences, they are more likely to

accept and incorporate this technology. This implies that businesses need to prioritize customer satisfaction to enhance acceptance and ensure the success of their service robots.

In addition, studies like Borghi and Mariani [60] delineated that service robots' excitement factor increases the satisfaction among users, leading to its acceptance in the tourism service sector. It was added that this satisfaction-acceptance relationship has led to the adoption of the tourism sector, including its automated service in the workforce. Leveraging on online reviews, Borghi et al. [61] have further deduced that customer satisfaction coming from the ease of use of service robots, appreciation and rapport build up, and being interactive have a correlation with overall adoption among hospitality industries. Seo and Lee [62] also implicated that there is a relationship between customer satisfaction and their acceptance of service robots as seen in their continuous utilization among restaurants and hotels. That is, people would continuously go back to the restaurants and hotels because of their adoption and acceptance of these service robots.

Wu and Huo [63] also presented the significant relationship between satisfaction and acceptance of service robots among hotels. It was explained that there was a positive satisfaction rate of service robots upon its introduction, and that its competence and perceived warmth service increased customers' acceptance. Chang et al. [64] implicated that there is a relationship between satisfaction and acceptance of Chinese consumers of the human-robot interaction. This is because several factors like anthropomorphism, performance efficacy, empathy, and motivation of using it.

Moreover, Han et al. [65] explored on the acceptance of hotel customers on service robots. Their results presented that a positive acceptance is seen when customers who are tech-savvy, sees the innovativeness of having service robots, attitude, and norms in using these technologies. Thus, easy to use and useful service robots would lead to an increased satisfaction and acceptance among consumers. Chen et al. [66] further reviewed and confirmed that service robots and the like have been widely accepted and adoption have been increasing since its establishment in 2008. However, there needs to be further studies since the review study have seen these relationships among developed countries, and among countries that have utilized service robots for a long time. Thus, it was hypothesized that:

H8. Customer satisfaction has a significant influence on customer acceptance.

3. Methodology

3.1. Participants

A purposive sampling approach was used in this study and collected 412 valid participants and gather the required information with written consent form (FM-RC-22-02-01). Participants in this study were Filipinos who had previously used robot kiosks or service robots, approved by the university ethics committee (FM-RC-22-01-01). As implicated by Buenaseda et al. [67], purposive sampling is ideal for assessing a target objective. It was added in the study that when rationale on needed assessment, especially for benchmark studies, purposive sampling technique is employed to properly align with the target assessment. This creates a representation of the data being analyzed [68]. In line with this study, assess the satisfaction, intention to use, and acceptance of robot kiosks and service robots. To ensure a comprehensive dataset, the survey was disseminated through various social media channels (obtaining 62 % of the respondents, 255), including Facebook, Reddit, and Instagram, as well as through in-person survey distribution (38 %, around 157 face-to-face surveys among fast-food and retail store customers) to respondents who had engaged with robot kiosks and service robots.

Table 1 summarizes the key demographic characteristics of the respondents. The gender distribution is nearly even, with (52.4 %) female and (47.4 %) male. Most participants are aged 18–24 years (55.1 %),

Table 1
Demographics characteristics.

Characteristics	Category	N	%
Gender	Male	196	47.4 %
	Female	215	52.4 %
	Prefer not to say	1	0.2 %
Age	18–24 years old	222	55.1 %
	25–34 years old	133	32.4 %
	35–44 years old	35	8.5 %
	45–64 years old	17	4.1 %
	65 years old and over	0	0 %
Education Attainment	Elementary	1	0.24 %
	Graduate	17	4.14 %
	Highschool	305	74.23 %
	Graduate	75	%
	College/	13	18.26 %
	Undergraduate		%
Occupational Status	College Graduate		3.16 %
	Graduate Degree		
	Student	214	52.10 %
	Employee/Self-employed	159	%
	Employer/Business	37	38.71 %
	Owner	1	%
Monthly Allowance/Income	Unemployed		8.99 %
	Less than 5,000 PHP	133	0.24 %
	5,001–10,000 PHP	87	32.4 %
	10,001–20,000 PHP	55	21.2 %
	20,001–30,000 PHP	59	13.4 %
	30,001–40,000 PHP	30	14.4 %
	40,001 PHP and higher	30	7.40 %
Living Environment		47	11.4 %
	Rural	194	47.2 %
Where do you usually see robotic kiosks? (Check all that applies)	Urban	217	52.8 %
	Retail Stores	64	15.6 %
	Fast Food	165	40.2 %
	Restaurants	173	42.1 %
	Both	9	2.20 %
Tech-savviness Perception	Others (service industries)		
	Very tech-savvy	226	56.0 %
	Somewhat tech-savvy	172	41.8 %
	Not very tech-savvy	9	2.20 %

followed by those aged 25–34 (32.4 %). A significant portion of respondents have a college-level education, with 74.23 % being college undergraduates and 18.26 % being college graduates. Romero and Lado [69] found that Generation Z, having grown up with fast-paced technology advancements quickly adapts to new technologies and expects them to be seamless and efficient. Their digital background and preference for convenience make them ideal for testing and refining new tech-related solutions, like self-service kiosks, making them the ideal participants.

In terms of occupation, (52.1 %) of respondents are students, while (38.71 %) are employed or self-employed. Most respondents (32.4 %) report a monthly income of less than 5,000 PHP. The living environments of the participants are almost equally divided, with (52.8 %) residing in urban areas and (47.2 %) in rural areas. Regarding tech-savviness, (56 %) of respondents consider themselves very tech-savvy, while (41.8 %) identify as somewhat tech-savvy. These individuals have experience with advanced technologies, making their feedback valuable for evaluating key features and appeal. Tech-savvy users are quickly embracing service robots due to convenience and technological progress. This growing use highlights the robots' practicality and potential impact [70]. By focusing on early adopters, developers can refine the technology, making it more accessible for less experienced users.

3.2. Questionnaire

The survey questionnaire encompassed 12 sections, reflecting the

latent variables of both the UTAUT model and the five dimensions of SERVQUAL. In the UTAUT model, constructs such as Effort Expectancy (EE), Facilitating Conditions (FC), and Social Influence (SI) each comprised three items, while Performance Expectancy (PE) included six items, totaling 18 indicators adapted from Alaiad and Zhou [40], Heerink [71], Mohammed [72], Seo [39], and Van Duy et al. [73]. The SERVQUAL dimensions included constructs such as Reliability (REL), Assurance (AS), Empathy (EMP), and Responsiveness (RES), each with four items, while Tangibles (TAN) consisted of six items, resulting in a total of 24 indicators adapted from Izogo and Ogba [74], Kharub et al. [75], and Kim and Lee [53]. Additionally, constructs such as Behavioral Intention (BI), Acceptance (AC), and Satisfaction (SAT) comprised four items each, adapted from Jin et al. [76], Seo [39], and Sung and Jeon [77]. Respondents were asked to provide their responses using a 5-point Likert scale, with options ranging from (1) strongly disagree to (5) strongly agree [78], to determine their perspectives on robot kiosks (supplementary file).

3.3. Structural equation modeling

Structural equation modeling (SEM) is a form of multivariate tools involving statistics for testing hypotheses about the relations between numerous variables [79]. Compared to other modeling approaches, SEM measures both the direct and indirect impacts on assumed causal relationships, enabling researchers to perform path analysis between observed and unobserved variables [80]. Sarstedt et al. [34] point out that higher-order constructs reduce complicated concepts into hierarchical structures when multiple path analysis is needed from the developed framework. It was added that higher-order constructs are now more commonly observed and utilized using partial least squares structural equation modeling (PLS-SEM), indicative of a growing trend in its application because of its intricate analysis method compared to lower-ordered SEM. This was implicated to be better used for integrated model or multiple pathed model since Woody [81] implicated that lower-ordered construct may reduce the significance level of variables far from the target output, while presence of mediating variable may also reduce the significance value.

In a study by German et al. [33], integrating a reflective higher-order SERVQUAL 5 dimension into the PLS-SEM framework enabled researchers to significantly evaluate how various service quality dimensions influenced consumers' intentions when choosing a package carrier. In another study, Huang [54] utilized UTAUT alongside PLS-SEM to assess users' behavior intentions toward technology adoption. Furthermore, PLS-SEM enabled researchers to assess the direct and indirect impacts of these measurements on behavioral intention and potential mediating effects—demonstrating greater sensitivity than its counterpart, the covariance-based SEM [82]. Therefore, PLS-SEM was employed in this study using SMART-PLS v4.0 to comprehensively evaluate the various relationships between variables.

4. Results

4.1. Result of initial SEM

In conducting the initial SEM analysis depicted in Fig. 3, it was found that 11 out of 12 hypotheses significantly supported the understanding of acceptance and satisfaction of robot kiosk. A dashed line was used to illustrate the non-significant relationship between behavioral intention and facilitating conditions, where the p-value exceeded 0.05. Reflecting with the study of Dash and Paul [82], the relationship should have p-values less than 0.05 to indicate significance. Additionally, measurement items with values less than 0.70 were considered insignificant. Hair et al. [83] discussed how these measurements should be removed to increase the model fit and implicate a final SEM.

Table 2 delineates the descriptive statistics, reliability, and factor loadings of the lower-order constructs. As indicated in the table, all final

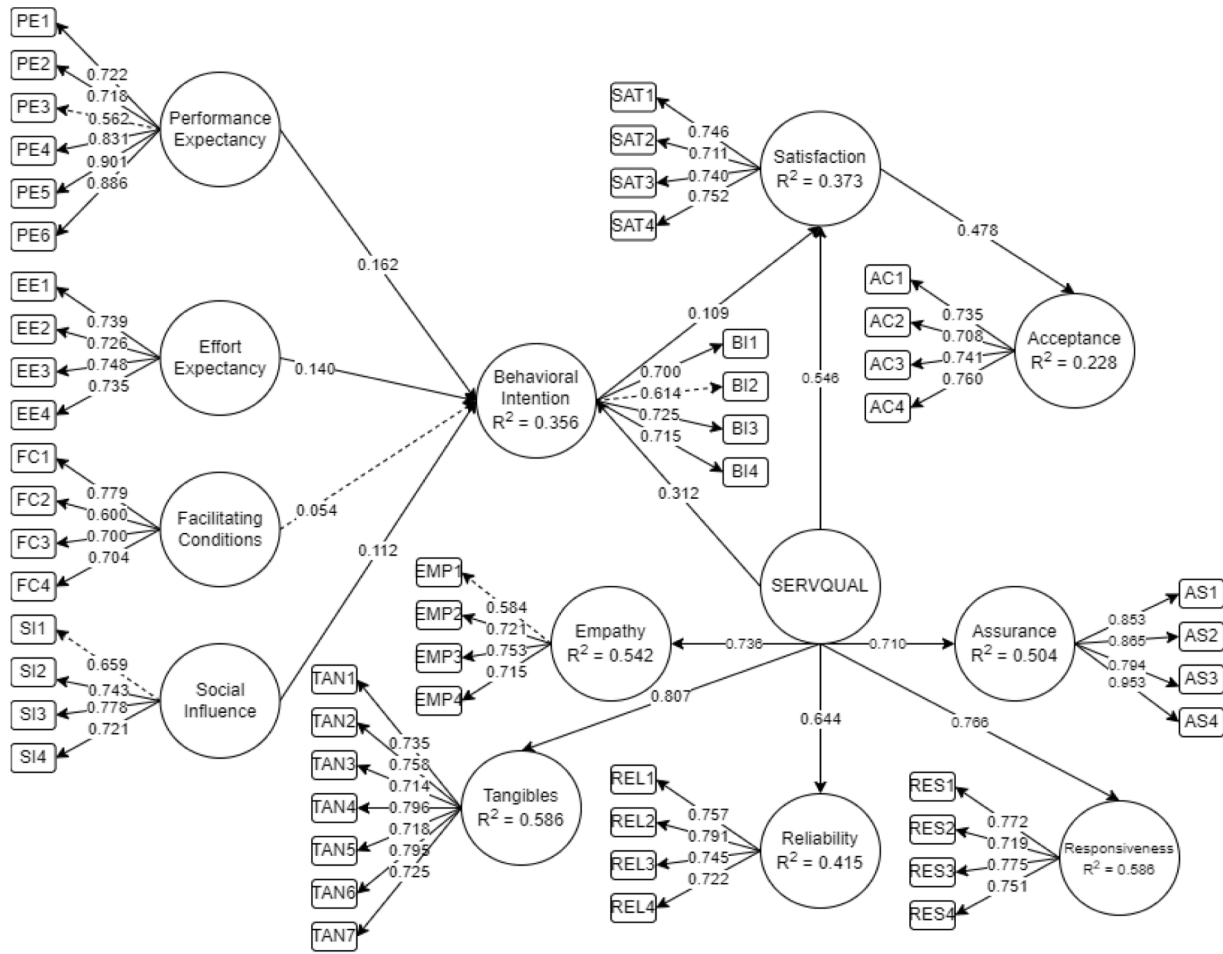


Fig. 3. Initial structural equation modeling.

factor loadings (FLs) surpassed the recommended threshold of 0.70 [82, 83]. This indicates that the items are sufficiently aligned with their respective constructs, demonstrating strong internal consistency.

In Table 3, the constructs exhibited solid reliability and validity, with both Cronbach's alpha (CA) and composite reliability (CR) values exceeding the 0.70 benchmark proposed by Hair [78] and German et al. [33]. Furthermore, the average variance extracted (AVE) for these constructs also met the recommended threshold of 0.50, confirming the constructs' ability to capture most of the variance in the measured items.

Discriminant validity was thoroughly assessed using the Fornell-Larcker Criterion (FLC) and the Heterotrait-Monotrait Ratio (HTMT), as detailed in Table 4. In accordance with the FLC threshold, the square root of AVE for each latent variable exceeded the correlation values with other variables. This confirmed that each construct is more closely related to its own indicators than to others [84]. Furthermore, the HTMT values were all below the recommended threshold of 0.85 [85], reinforcing the distinctiveness of the constructs and thereby affirming the discriminant validity of the measurement model. These results collectively validate the model's structure and ensure the constructs' accurate representation.

Table 5 provides an analysis of the model fit using all the parameter estimates. The Standardized Root Mean Square Residual (SRMR) is reported at 0.050, which is within the acceptable threshold of 0.08, indicating an adequate fit of the model to the data. The Chi-square value is 2.240, which exceeds the recommended maximum value of 5.00. Moreover, the Normed Fit Index (NFI) value of 0.876 is above the minimum threshold of 0.8, indicating a reasonably good fit. Collectively, these results suggest that the model demonstrates an overall acceptable fit based on the indices analyzed.

In addition, Table 6 summarizes the results of hypothesis tests examining various factors influencing Behavioral Intention (BI) in the context of robot kiosks, using the UTAUT and SERVQUAL models. Performance Expectancy (PE) significantly influences BI ($\beta = 0.162, p = 0.005$), indicating that users are more likely to use the kiosks when they expect them to perform well. Similarly, Effort Expectancy (EE) positively affects BI ($\beta = 0.156, p = 0.004$), while Social Influence (SI) also has a positive impact ($\beta = 0.112, p = 0.016$), suggesting that ease of use and social factors drive users' intentions.

However, Facilitating Conditions (FC) do not significantly impact BI ($\beta = 0.054, p = 0.283$). For the SERVQUAL model, it shows that overall service quality significantly influences BI ($\beta = 0.325, p < 0.001$) and Satisfaction (SAT) ($\beta = 0.549, p < 0.001$) – highlighting the importance of perceived service quality in shaping user intentions and satisfaction. Furthermore, BI contributes to SAT ($\beta = 0.104, p = 0.042$), and SAT significantly enhances Assurance (AC) ($\beta = 0.478, p < 0.001$). The higher-order SERVQUAL dimensions—Assurance (AS), Empathy (EMP), Reliability (REL), Responsiveness (RES), and Tangibles (AN)—all show strong positive effects (β ranging from 0.646 to 0.807, $p < 0.001$), reinforcing their importance as key contributors to overall service quality perceptions.

The final SEM in Fig. 4 illustrates the relationships between various factors influencing satisfaction, user behavior, and acceptance. The model shows that performance expectancy, effort expectancy, and social influence all significantly impact behavioral intention. Behavioral intention, in turn, affects satisfaction, which influences the acceptance of services. Additionally, SERVQUAL plays a central role in positively impacting various service dimensions, including empathy, tangibles, reliability, assurance, and responsiveness, all of which contribute to

Table 2
Statistical analysis of indicators.

Variable	Item	Mean	SD	Factor Loading	
				Initial	Final
Performance Expectancy	PE1	4.163	0.624	0.722	0.749
	PE2	4.032	0.733	0.718	0.725
	PE3	3.830	0.882	0.562	–
	PE4	4.263	0.704	0.831	0.739
	PE5	4.246	0.735	0.901	0.831
	PE6	4.251	0.703	0.886	0.902
Effort Expectancy	EE1	4.253	0.718	0.739	0.755
	EE2	4.161	0.703	0.726	0.721
	EE3	4.217	0.754	0.748	0.738
	EE4	4.292	0.706	0.735	0.736
Facilitating Conditions	FC1	4.071	0.794	0.779	–
	FC2	3.720	0.815	0.600	–
	FC3	3.832	0.842	0.700	–
	FC4	3.813	0.955	0.704	–
Social Influence	SI1	3.971	0.792	0.659	–
	SI2	3.730	0.841	0.743	0.711
	SI3	3.871	0.877	0.778	0.814
	SI4	4.187	0.793	0.721	0.795
Behavioral Intention	BI1	4.253	0.728	0.700	0.701
	BI2	4.044	0.827	0.614	–
	BI3	4.229	0.802	0.725	0.745
	BI4	4.394	0.680	0.715	0.771
Acceptance	AC1	4.129	0.722	0.735	0.735
	AC2	4.083	0.724	0.708	0.708
	AC3	4.144	0.778	0.741	0.741
	AC4	4.243	0.700	0.760	0.760
Satisfaction	SAT1	4.170	0.738	0.746	0.745
	SAT2	4.088	0.689	0.711	0.710
	SAT3	4.131	0.730	0.740	0.741
	SAT4	4.217	0.704	0.752	0.754
Reliability	REL1	3.903	0.796	0.757	0.757
	REL2	3.749	0.779	0.791	0.79
	REL3	3.766	0.885	0.745	0.745
	REL4	3.881	0.827	0.722	0.722
Assurance	AS1	4.112	0.778	0.853	0.853
	AS2	4.007	0.689	0.865	0.866
	AS3	4.007	0.824	0.794	0.795
	AS4	4.102	0.834	0.953	0.951
Empathy	EMP1	4.005	0.816	0.584	–
	EMP2	4.012	0.692	0.721	0.717
	EMP3	3.988	0.875	0.753	0.735
	EMP4	4.224	0.766	0.715	0.776
Responsiveness	RES1	4.173	0.733	0.772	0.773
	RES2	4.012	0.759	0.719	0.72
	RES3	4.107	0.783	0.775	0.774
	RES4	4.287	0.715	0.751	0.750
Tangibles	TAN1	4.217	0.734	0.735	0.735
	TAN2	4.051	0.697	0.758	0.758
	TAN3	4.136	0.787	0.714	0.714
	TAN4	4.319	0.724	0.796	0.796
	TAN5	4.311	0.687	0.718	0.715
	TAN6	4.338	0.654	0.795	0.795
	TAN7	4.338	0.701	0.725	0.725

Table 3
Convergent validity.

Variables	Cronbach's Alpha	Composite Reliability	Average Variance Extracted (AVE)
Acceptance	0.775	0.826	0.542
Assurance	0.785	0.924	0.753
Behavioral Intention	0.884	0.783	0.547
Effort Expectancy	0.877	0.827	0.544
Empathy	0.796	0.787	0.552
Performance Expectancy	0.762	0.893	0.627
Reliability	0.706	0.840	0.568
Responsiveness	0.765	0.841	0.569
Satisfaction	0.775	0.827	0.544
Social Influence	0.774	0.818	0.600
Tangibles	0.796	0.899	0.561

overall service quality. This model highlights the interconnected nature of these variables in shaping user perceptions and intentions.

5. Discussion

As the industry rapidly evolves, the food and service sectors are increasingly turning to new technologies to improve customer engagement and enhance service efficiency. This shift has led to the introduction of robot kiosks, which are transforming the way customers interact with businesses. Despite this, ongoing discussions continue about the actual benefits of these kiosks and their impact on customer satisfaction within the digitalized service industry [21]. This study analyzed the key factors influencing the intention to adopt robot kiosks in selected food retail and service stores. By integrating the UTAUT model with a service quality approach, it evaluates how various factors affect customers' willingness to adopt this technology.

Findings presented that service quality emerged as a significant factor affecting satisfaction ($\beta = 0.549, p < 0.001$). It can be inferred that consumers are satisfied with the services provided by robot kiosks, as these kiosks effectively meet their needs and improve the efficiency and reliability of the service. This satisfaction arises from the perception that robot kiosks enhance the overall customer experience. This finding aligns with Lukanova and Ilieva [89] who observed that robot kiosks improve the ordering and payment process, making it faster and more convenient for both customers and businesses. Similarly, Chiang and Trimi [90] noted that while robot kiosks are generally reliable and efficient, customer satisfaction is primarily driven by their ability to provide consistent and trustworthy service. These insights offer valuable guidance for managers seeking to implement robot kiosks effectively, highlighting the importance of delivering consistent and reliable service. Furthermore, Kim and Lee [53] emphasized that consumers are more likely to remain satisfied and continue using a robot kiosk if it proves genuinely helpful in enhancing their experience or simplifying tasks. In this context, consumers prioritize a well-functioning system that is stable, reliable, and responsive over other aspects, such as appearance or motion.

In the analysis of higher-order constructs, the tangible variable was identified as having the most pronounced effect on behavioral intention towards robot kiosks, followed by empathy, responsiveness, and assurance having progressively lesser effects, and reliability showing the least significant impact. Tangibles demonstrated the most potent effect on behavioral intention ($\beta = 0.807, p < 0.001$). In this study, it was found that tangible aspects of robot kiosks such as appealing design, appropriate weight, optimal size, pleasing color, and neatness contribute to enhancing the overall aesthetics of the service market. These factors are closely aligned with significantly influencing consumer intention and satisfaction. This observation is validated by Brengman et al. [21], who highlighted that the notable performance of the humanoid service robots (HSRs) indicates that its tangible features such as design, neatness, color, and interactive capabilities play a crucial role in effectively engaging consumers. These tangible elements help attract attention, captivate passersby, and convert them into customers. Furthermore, Seyitoğlu and Ivanov [91] emphasized that robots introduce a novel and entertaining element to the dining experience due to their engaging physical appearance and attraction. While service robots can enhance this experience, it is also essential to ensure that food quality and ambiance meet customer expectations to fully complement the appeal of robotic service and maintain overall service quality. Moreover, Wang et al. [92] found that the appearance of service robots plays a crucial role in consumer acceptance. Robots with unattractive designs tend to discourage users from interacting with them. Therefore, service robots need to feature appealing and user-friendly designs to maximize their acceptance and usage in service settings as tangible domains.

In relation to the preceding variable, empathy ($\beta = 0.782, p < 0.001$) exhibited the second most substantial effect on service quality. This study found that empathy significantly enhances service quality, as

Table 4
Discriminant validity.

FLC											
Construct	AC	AS	BI	EE	EMP	PE	REL	RES	SAT	SI	TAN
AC	0.736										
AS	0.645	0.868									
BI	0.538	0.667	0.740								
EE	0.677	0.731	0.613	0.738							
EMP	0.726	0.706	0.686	0.704	0.743						
PE	0.643	0.765	0.723	0.646	0.617	0.792					
REL	0.718	0.736	0.696	0.706	0.686	0.666	0.754				
RES	0.722	0.627	0.707	0.636	0.664	0.67	0.748	0.755			
SAT	0.678	0.748	0.721	0.541	0.705	0.732	0.688	0.713	0.738		
SI	0.636	0.758	0.658	0.685	0.725	0.717	0.614	0.602	0.682	0.775	
TAN	0.713	0.747	0.572	0.713	0.644	0.648	0.748	0.658	0.731	0.686	0.749
HTMT ratio											
Construct	AC	AS	BI	EE	EMP	PE	REL	RES	SAT	SI	TAN
AC											
AS	0.691										
BI	0.695	0.734									
EE	0.558	0.525	0.655								
EMP	0.748	0.669	0.671	0.471							
PE	0.679	0.592	0.715	0.725	0.500						
REL	0.608	0.588	0.761	0.504	0.424	0.580					
RES	0.622	0.689	0.662	0.536	0.729	0.634	0.507				
SAT	0.769	0.771	0.684	0.659	0.680	0.818	0.761	0.668			
SI	0.517	0.771	0.555	0.615	0.787	0.610	0.655	0.743	0.549		
TAN	0.745	0.682	0.582	0.506	0.676	0.577	0.542	0.732	0.711	0.402	

Table 5
Model fit analysis.

Measures	Parameter Estimates	Minimum Cut-off	References
SRMR	0.050	<0.08	Hu and Bentler [86]
Chi-square	2.240	<5.00	Hooper et al. [87]
NFI	0.876	>0.90	Baumgartner and Homburg [88]

Table 6
Summarized results.

Hypothesis	Relationship	Beta	P-value	Decision
1	PE → BI	0.162	0.005	supported
2	EE → BI	0.156	0.004	supported
3	FC → BI	0.054	0.283	not supported
4	SI → BI	0.112	0.016	supported
5	SERVQUAL → BI	0.325	<0.001	supported
6	SERVQUAL → SAT	0.549	<0.001	supported
7	BI → SAT	0.104	0.042	supported
8	SAT → AC	0.478	<0.001	supported
Higher order	SERVQUAL → AS	0.711	<0.001	supported
Higher order	SERVQUAL → EMP	0.782	<0.001	supported
Higher order	SERVQUAL → REL	0.646	<0.001	supported
Higher order	SERVQUAL → RES	0.765	<0.001	supported
Higher order	SERVQUAL → TAN	0.807	<0.001	supported

robot kiosks were consistently available, understood consumer needs, and responded effectively. Additionally, by offering personalized assistance and prioritizing customers' best interests, these kiosks helped establish a positive perception of the service experience between providers and customers. The study by Chiang et al. [93] discussed how these human-like features can enhance the intimacy between customers and robots, positively influencing perceived service quality. These features allow robots to interact more efficiently with customers, providing a service that feels more personalized and emotionally attuned to individual needs. Moreover, Kharub et al. [75] added that empathy significantly influences the emotional connection between customers and

social robots. The study emphasized that empathetic interactions not only enhance personal engagement but also have a significant impact on the service quality delivered by social robots. This emotional engagement, driven by empathy, is a key factor in customer satisfaction and increases the likelihood of continued service use. When robots exhibit an understanding of and responsiveness to consumers' needs, it improves the overall service quality. This indicates that integrating empathetic behaviors into robot interactions is crucial for fulfilling service quality standards [94].

Moreover, responsiveness was perceived to have the third highest effect on higher-order constructs ($\beta = 0.765, p < 0.001$). As indicated by this study, it was determined that consumers have confidence in robot kiosks, believing they are both willing and prepared to assist clients efficiently, and making them good assistants. This signifies that when robots exhibit high levels of responsiveness, they are better able to address consumers' needs, leading to an improved overall service experience. According to Zhu [45], when customers perceive a high level of responsiveness in robot kiosks, it not only enhances their service experience but also strengthens their intention to reuse these services. This relationship underscores the critical role of responsiveness in shaping overall service quality. Similarly, Marinakou et al. [95] found that when the service quality of service robots is high, particularly in terms of responsiveness, customers tend to have more positive experiences and are more likely to continue using the technology. In fast-food restaurants, service robots represent a technological advancement aimed at improving speed, convenience, and overall food ordering experience. In addition, responsiveness plays a key role in determining service quality, significantly boosting customer satisfaction. This improvement is largely due to the prompt and efficient service these systems provide, which helps create a smooth and satisfying customer experience [96].

On the other hand, assurance ($\beta = 0.711, p < 0.001$) was found to impact service quality significantly. Based on the constructs, it was seen in the assurance measurements that robot kiosks increase the trust of consumer, offer friendly services, security, and equipped with knowledge that answers customer inquiries. This indicates that a high level of assurance in service robots considerably enhances the overall quality of the service provided. This finding is supported by Choi et al. [97], who

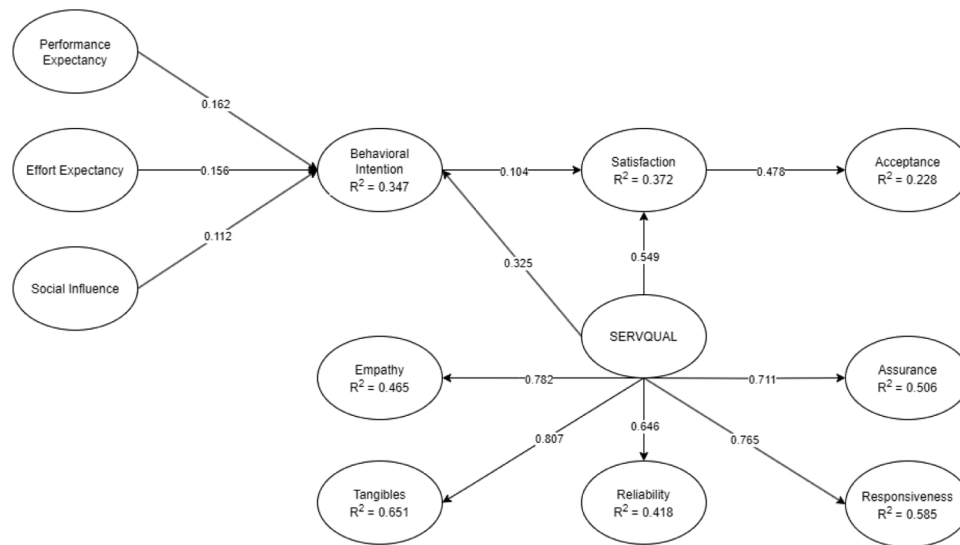


Fig. 4. Final structural equation modeling.

observed that integrating service robots is increasingly recognized as a key element of the future workforce. They emphasized that high-quality service, which includes adequate security and trust, is crucial for achieving customer satisfaction, especially in the hotel industry. Moreover, Seo and Lee [62] highlighted the importance of consumer assurance in the service industry, noting that when consumers trust in the knowledge and service effectiveness of robots, their overall experience improves significantly. This reassurance leads to a more positive perception of the service, resulting in higher satisfaction with robot-assisted interactions.

The least significant factor in higher-order constructs is reliability ($\beta = 0.646, p < 0.001$), which measures the consistency and dependability of the service provided by automated systems. As shown by this study, all measured items were found to be significant, including addressing client issues, meeting customer expectations, completing tasks on the first attempt, and delivering services within the expected timeframe, all of which contribute to better service quality. As service robots are not yet widely used, reliability might seem less important as people are still getting accustomed to and building trust in these systems. However, as robots become more common, reliability will likely become increasingly crucial. Saeheng [98] highlighted that meeting expectation is crucial in shaping customer perceptions of quality by ensuring consistent dependability and alignment with user expectations, especially in self-service environments. Nguyen et al. [99] similarly emphasize that executing task on time is essential for building trust and confidence in robotic kiosks. Improved reliability often leads to increased customer satisfaction and a greater sense of value. Mandić et al. [100] further support this by noting that high reliability significantly enhances customer perceptions of service quality, leading to greater satisfaction. Their study underscores that a robot kiosk, which reliably performs its tasks, can greatly enhance customer satisfaction compared to traditional service methods that heavily rely on human involvement. This consistent emphasis on reliability across studies underscores its crucial role in improving service quality and customer satisfaction.

Results indicate that customer satisfaction is crucial for the acceptance of robot kiosks ($\beta = 0.478, p < 0.001$), significantly influencing users' likelihood to adopt and continue using these automated systems. This study suggests that satisfied consumers are more likely to embrace and continue using robot kiosks, highlighting the crucial role of satisfaction. Key factors include meeting customer needs and preferences, ensuring a satisfying service experience, and enhancing efficiency and reliability, all of which contribute to increased satisfaction and the widespread acceptance and sustained usage of robot kiosks. Sharma

et al. [101] demonstrated that when service robots meet or exceed customer expectations, satisfaction increases, boosting the willingness to accept these robots as viable alternatives to traditional human-delivered services. Similarly, Luo et al. [102] found that age affects the adoption and acceptance of service robots. Their research showed that younger users adapt quickly to robotic kiosks, whereas older adults often require additional support and greater familiarity with the technology to improve their acceptance levels. In this study, 222 respondents aged 18 to 24, representing 55.1 % of the sample, highlight the significant impact of satisfaction on the acceptance of robot kiosks. Additionally, Min et al. [103] explored the behavioral intentions of individuals aged 40 to 60 toward self-ordering kiosks, revealing that this older demographic may exhibit different behavioral intentions. The study, which included 52 respondents in this age range, suggests that addressing the specific needs and preferences of older adults could significantly enhance their acceptance and use of these technologies.

As for the UTAUT domains, it was found that three out of four constructs were significant to behavioral intention. Performance expectancy was determined to have the highest impact ($\beta = 0.162, p < 0.001$), followed by effort expectancy ($\beta = 0.156, p < 0.001$), and social influence ($\beta = 0.112, p < 0.001$). However, facilitating conditions was perceived to be an insignificant factor ($\beta = 0.054, p < 0.283$). This study revealed that the service market in the Philippines is currently facing limitations that hinders the effectiveness of utilizing robot kiosk. The findings indicate that some consumers lack the necessary knowledge to operate robot kiosks, the service market lacks sufficient resources to support their use, and there is no staff or personal guidance provided to assist with operating the service kiosks.

Ivanov et al. [20] explored the difficulties encountered by businesses in sectors like hotels, airports, and restaurants when integrating robot kiosks. The research highlighted that, despite progress in social robotics, the full potential of these technologies remains untapped because of limitations in the local service market. The study by Roh et al. [104] revealed that while facilitating conditions significantly influence the intention to use technology, their impact is also indirect, working through the user's attitude. This implies that facilitating conditions might not directly determine the intention to use but instead affect it by first cultivating a positive attitude. This underscores the importance of establishing a supportive environment that nurtures positive attitudes toward technology, ultimately leading to greater adoption and usage.

Moreover, the study revealed that performance expectancy impacts the intention to use customer-service robots. In this study, it was

suggested that service robots are seen as valuable tools for improving how efficiently customers are served, leading to better productivity and quicker service times within the industry. Also, people are more likely to choose robot kiosks because they value faster transactions, which makes their overall experience smoother and more convenient. This relationship underscores that the perceived benefits and efficiency of a technology play a crucial role in motivating users to adopt and utilize it. Osakwe et al. [105] found that, for Generation Z shoppers, performance expectancy was a key factor in their willingness to interact with customer service robots. Similarly, although older adults were less likely to expect enjoyment from social robots compared to younger individuals, they still showed a slight preference for accepting them. This indicates that performance expectancy positively influences the adoption of social robots [106]. Additionally, 226 respondents from this study, or 56 % of the sample, considered themselves very tech-savvy. This finding aligns with Zhang et al. [107], which suggests that familiarity with advanced technological devices, such as artificial intelligence and kiosks, enhances users' expectations of a robot's performance and, consequently, increases their intention to use the robot.

Effort expectancy is a significant determinant of behavioral intentions. In this research, effort expectancy was evaluated through items such as ease of use, clarity of robot instructions, ease of becoming proficient with robot kiosks, and the simplicity of navigating them. These factors were found to significantly influence consumer intention and satisfaction. The study by Guggemos et al. [108], which employed the UTAUT model to evaluate service robots in a different industry, demonstrated that the impact of effort expectancy on behavioral intentions is nuanced and influenced by factors such as ease of use and prior experience with robots. The findings emphasized that when a service robot or kiosk is perceived as user-friendly and requires minimal effort to operate, users are more likely to adopt it. Similarly, the findings of Alaiad and Zhou [40] highlighted the significance of effort expectancy in shaping behavioral intentions. They measured factors such as the clarity of system inputs, the complexity of interactions, the perceived ease of achieving outcomes, and the overall effort required. These elements underscore how effort expectancy influences users' willingness to engage with the system.

Under the UTAUT model, social influence plays a crucial role in shaping the intention to use a robot kiosk. This study found that social influence factors—such as significant people in one's life advocating for the use of robot kiosks to enhance productivity, valued opinions suggesting their usefulness, and the belief that using them would create a positive impression within the community—were significant in influencing social behavior. This implies that when individuals observe others using and endorsing robot kiosks, it strongly shapes their perceptions and increases the likelihood of adoption. Supporting this, Yeoh and Chin [26] demonstrated that social influence significantly impacts consumer intentions to use service robots in Malaysia. Additionally, Tojib et al. [109] found that social influence heightens customers' perceptions of the challenges involved in using a service robot, underscoring the importance of incorporating social influence into strategies for promoting robot adoption. By leveraging social influence, companies can develop more effective strategies to encourage customers to adopt service robots, ultimately enhancing their return on investment. Furthermore, Lu et al. [110] highlighted that while social influence does affect technology adoption, its impact varies depending on the context. Moreover, social influence is more significant in mandatory situations, such as in workplaces where compliance is required, and less so in voluntary contexts where individuals have more freedom to make their own choices.

The results indicated a significant influence of behavioral intention on satisfaction ($\beta = 0.104$, $p < 0.042$). The study observed that consumers are likely to continue using the robot kiosk and anticipate using service robots more frequently in the future. This suggests that the stronger the intention to use the kiosk, the more likely individuals are to feel satisfied with it. Building on this, Çallı et al. [111] found that

consumers' perceived value of service robots is based not only on their current capabilities but also on expectations for more frequent engagement in the future. Additionally, Jia et al. [112] determined that the quality of interaction with robot kiosks plays a crucial role in user satisfaction within the hospitality industry. Aspects such as the robot's operation and overall user experience significantly influence users' intentions to use the kiosk, which in turn impacts their satisfaction with the service. Gelbrich et al. [113] also observed that interactions between humans and machines can enhance customer satisfaction. Specifically, digital assistants that provide warm and reassuring feedback significantly boost satisfaction and persistence. The study implied that frequent use of digital assistants can lead to greater contentment among users.

Comparing the output among related studies, a summarized presentation of the current findings against recent literatures are provided (Table 7).

5.1. Theoretical implications

To fully understand the adoption of robot kiosks in retail stores, it is crucial to apply a robust theoretical framework that integrates various influencing factors. This study combined the Unified Theory of Acceptance and Use of Technology (UTAUT) with the Service Quality (SERVQUAL) model to explore the adoption of robot kiosks in retail settings. The UTAUT model, helps explain how users' perceptions of these factors drive their behavioral intention to adopt new technologies. Meanwhile, the SERVQUAL model, sheds light on how service quality impacts user satisfaction and technology acceptance.

By integrating these two frameworks, this study provides a comprehensive understanding of the variables affecting the acceptance and use of robot kiosks. While the UTAUT model highlights the importance of perceived usefulness and ease of use, the SERVQUAL model emphasizes how the quality-of-service delivery influences user satisfaction. This dual approach not only broadens the theoretical perspective on technology adoption but also addresses practical aspects of service quality in the context of robot kiosks. This integrated theoretical approach offers valuable insights for businesses and policymakers, guiding the effective deployment of robot kiosks and advancing the use of AI technologies in the retail sector.

5.2. Practical and managerial implications

5.2.1. Managerial constraints

To ensure that robot kiosks are well-received and effective in retail environments, companies must tackle several important managerial challenges, including manufacturability, social, ethical, and economic factors which are typically constraints. Manufacturability as a constraint is a key challenge in deploying robot kiosks. Burke [120] assessed that industries need to develop kiosks that are not just technologically advanced, but also reliable and affordable. This means working closely with manufacturers to create designs that can be produced on a large scale and easily integrated into existing retail systems. Managing manufacturability effectively can lead to smoother implementation, lower maintenance costs, and improved operational efficiency.

Social factors are also crucial. Introducing automated systems like robot kiosks requires careful consideration of the social dynamics within retail environments. Bitner et al. [121] stated that companies need to address employee concerns about job displacement and ensure that these kiosks enhance, rather than diminish customer interactions. By facilitating positive interactions between technology and customers and providing proper training for staff, companies can boost both employee morale and customer satisfaction – delineating the social constraints.

In addition, ethical considerations are vital for building trust and ensuring the responsible use of technology. Bonarini [122] indicated that, service companies must tackle issues related to data privacy and algorithmic fairness to avoid biases and safeguard consumer

Table 7
Comparative results and implications.

Current Findings	Literatures
Service Quality domains affected satisfaction. Tangibles as the highest impactful variable – appealing design, appropriate weight, optimal size, pleasing color, and neatness – the overall aesthetic prompted the significant effect.	Support: This observation is validated by Brengman et al. [21] and Seyitoğlu and Ivanov [91], who highlighted that the notable performance of the humanoid service robots (HSRs) indicates that its tangible features such as design, neatness, color, and interactive capabilities play a crucial role in effectively engaging consumers. These tangible elements help attract attention, captivate passersby, and convert them into customers. Contrasting: Unappealing and under-designed robot kiosks and service robot may lead to negative impact on the overall acceptance of users [92]. To which, Chiang and Trimi [90] presented tangibles to be insignificant.
Empathy is significant – being consistently available, understood consumer needs, and responded effectively.	Support: Chiang et al. [93], Kharub et al. [75], and De Kervenoael et al. [94] discussed how these human-like features can enhance the intimacy between customers and robots, positively influencing perceived service quality. Contrasting: The implementation of full empathy factor among robot kiosks and service robots are limited [114]. There needs to have further improvement for a holistic emotional touch of the current AI-enabled service robots. Similarly, Chiang and Trimi [90] implicated that empathy is not a significant variable.
Responsiveness is significant – confidence in robot kiosks, believing they are both willing and prepared to assist clients efficiently, and making them good assistants.	Support: An et al. [96], Marinakou et al. [95], and Zhu [45] found that when the service quality of service robots is high, particularly in terms of responsiveness, customers tend to have more positive experiences and are more likely to continue using the technology. Contrasting: Chang and Trimi [90] implicated that it is difficult for robots to independently perform certain tasks. This may lead to lower responsiveness implications among users. Therefore, proper AI implementation is needed to enhance responsiveness.
Assurance is significant – increase the trust of consumer, offer friendly services, security, and equipped with knowledge that answers customer inquiries.	Support: Seo and Lee [62] highlighted the importance of consumer assurance in the service industry, noting that when consumers trust in the knowledge and service effectiveness of robots, their overall experience improves. Contrasting: Utilitarian values are more prominent among service robots, as evidenced by Odekerken-Schroder et al. [115]. There needs to have an exchange of user and functionality to acquire assurance among these robots for satisfaction to be achieved.
Reliability is significant – addressing client issues, meeting customer expectations, completing tasks on the first attempt, and delivering services within the expected timeframe	Support: Mandić et al. [100] and Nguyen et al. [99] emphasized that executing task on time is essential for building trust and confidence in robotic kiosks – building reliability. Contrasting: There are still service failure considered and thus, reliability may not at all be significant. This is because users are now forced to apply and utilize the robots due to availability despite not being fully embedded with proper deployment [116].
Satisfaction was achieved among robot kiosks and service robots since customer needs and preferences are met, ensuring a satisfying service	Support: Lou et al. [102], Luo et al. [102], and Sharma et al. [101] demonstrated that when service robots meet or exceed customer expectations, satisfaction increases, boosting the willingness to

Table 7 (continued)

Current Findings	Literatures
experience, and enhancing efficiency and reliability	accept these robots as viable alternatives to traditional human-delivered services. Contrasting: Borghi and Mariani [60] implicated that users do not have moderating effect with satisfaction and performance of these robots. It was explained that there should be an excitement factor to enhance overall acceptance. Nonetheless, it could be posited that there is a significant effect but was limited only among online reviews and may not fully translate to actual utilization.
Performance expectancy is seen to be significant because it is seen as valuable tools for improving how efficiently customers are served, leading to better productivity and quicker service times within the industry.	Support: Similar output were seen, in accordance, older adults were less likely to expect enjoyment from social robots compared to younger individuals, they still showed a slight preference for accepting them. This indicates that performance expectancy positively influences the adoption of social robots [105,106]. Contrasting: Despite being widely common and evidently present in the market, most people would still have a hard time understanding and utilizing service robots, leading to low acceptance [117]. Nonetheless, when tech-savvy individuals like the ones in this study have adopted and highly utilized these robots, there is satisfaction and positive acceptance.
Effort expectancy is seen to have contributed to the intention to use and accept the robot kiosks and service robots because the ease of use, clarity of robot instructions, ease of becoming proficient with robot kiosks, and the simplicity of navigating them.	Support: Guggemos et al. [108] and Soderlund [117] showed similar output, effort expectancy is a significant determinant of satisfaction and acceptance for robot kiosks and service robots. Contrasting: Lin and Lee [118] established that effort expectancy has no significant influence on adoption. It would only be accepted and widely utilized when security and trust is established. Nonetheless, this study's output provided that effort expectancy was significant since only service is being provided, manipulation of orders and demands, and no personal information is needed for the robots to function – leading to a positive effect of effort expectancy in this study.
Social influence is also significant as others in the service industry are also using it and that the community overall is accepting of having it among service industries.	Support: Tojib et al. [109] and Yeoh and Chin [26] found that social influence heightens customers' perceptions of the challenges involved in using a service robot, underscoring the importance of incorporating social influence into strategies for promoting robot adoption. Contrasting: Lajante et al. [119] provided evidence on when service robot is not satisfying – showing that users with lower social-emotional state have correlated negative emotions. Moreover, there is only an instrumental support for these robots that may lead to positive interaction among users. Overall adoption is needed for the acceptance of the general public.
It was discovered that some consumers lack the necessary knowledge to operate robot kiosks, the service market lacks sufficient resources to support their use, and there is no staff or personal guidance provided to assist with operating the service kiosks, leading to the insignificant effect of facilitating conditions.	Support: Ivanov et al. [20] explored the difficulties encountered by businesses in sectors like hotels, airports, and restaurants when integrating robot kiosks. The research highlighted that, despite progress in social robotics, the full potential of these technologies remains untapped because of limitations in the local service market. Roh et al. [104] revealed that while facilitating conditions significantly influence the intention to use technology, their impact is also indirect,

(continued on next page)

Table 7 (continued)

Current Findings	Literatures
	working through the user's attitude. Contrasting: Malls and other service industries like fast-food restaurants have deployed more robot kiosks and service robots, enhancing the overall performance for user utilization in the current market [16]. These, however, have yet reflected as seen in the current findings.

information. Transparent data protection practices and ethical guidelines not only enhance the company's reputation but also help avoid potential legal and regulatory issues, supporting long-term success. By addressing these managerial challenges, companies can make the most of robot kiosks to enhance service delivery, drive technological progress, and gain a competitive edge. A thoughtful approach to manufacturability, social integration, ethical practices, and economic feasibility will ensure that robot kiosks contribute positively to the retail industry.

5.2.2. Government agencies

This study provided valuable insights into the factors influencing the adoption of robot kiosks across various service sectors, which can inform government policies and initiatives. The findings highlight the importance of integrating the UTAUT model and the SERVQUAL 5 dimensions to understand how different factors impact the acceptance and satisfaction associated with robot kiosks. Government agencies involved in technological advancement and industry regulation can leverage this information to develop comprehensive standards that address these rapid technological developments.

Supporting infrastructure development is crucial for facilitating the deployment of robot kiosks. This includes improving digital connectivity and technological resources to ensure that kiosks can be effectively implemented across diverse regions. Additionally, establishing educational initiatives and training programs focused on the benefits and practical use of robot kiosks can enhance understanding and acceptance among businesses and consumers.

5.3. Limitations and future research

While this study provided valuable insights into the factors affecting the adoption of robot kiosks in retail stores, it has limitations that suggest areas for further exploration. First, the study relied on the UTAUT framework combined with service quality to understand consumer behavior. Future research could benefit from integrating additional models, such as the Technology Acceptance Model (TAM) or the Theory of Planned Behavior (TPB), to offer alternative perspectives. Second, the study focused solely on current users of robot kiosks, excluding non-users who might favor traditional service methods. Including non-users in future research could yield a more complete understanding of adoption dynamics. Moreover, older generation users who may not be well adopted with technologies, lower technology self-efficacy individuals, and have lesser interaction with technology may provide other findings that future researchers could consider. Third, while behavioral intentions were examined, actual usage behavior was not explored. Future studies could address this gap to see how intentions translate into real-world actions since this was a benchmark study and only current users were considered. Additionally, expanding research to different cultures or demographic groups could reveal diverse attitudes toward robot kiosks or service robots. Finally, although this study employed structural equation modeling, future research could utilize advanced techniques like machine learning to enhance the precision and depth of the findings. By addressing these areas, future research can build on this study and provide more comprehensive insights into the adoption of robot kiosks in retail settings.

6. Conclusion

The integration of robot kiosks and service robots across various service sectors in the Philippines represents a significant technological advancement, profoundly impacting industries such as retail, hospitality, and healthcare. These robots are transforming service delivery by automating tasks, enhancing operational efficiency, and enriching customer engagement. However, the deployment and availability has yet been assessed in the country. Based on related literatures, there are limited studies, especially among developing countries like the Philippines that holistically assessed the service quality, satisfaction, and overall acceptance of users for the service robots and robot kiosks. The main results and implications are summarized:

a. Service Quality Domains:

From the RATER dimensions, tangibles, empathy, responsiveness, reliability, and assurance were all found to be significant variables affecting consumers' satisfaction and intention to use service robots and robot kiosks. That is, the design and interactive capabilities of the robots have the most significant impact on customers' intentions to use these technologies. Meanwhile, creating a more personalized and effective service experience, as well as consistency of service, responds effectively, especially to the needs of consumers, security, trust, and meeting customer expectations further enhanced satisfaction.

b. Technology Acceptance Factors:

Considering the UTAUT domains, performance expectancy, effort expectancy, and social influence are key factors influencing customers' intention to adopt robot kiosks and service robots. Performance expectancy, reflecting the perceived benefits and efficiency of the technology, emerged as the most critical determinant. This is followed by effort expectancy and social influence, wherein ease of use, clarity in utilization, simplicity, and overall herd adoption together shaped customer perceptions and motivation to use robot kiosks – leading to the acceptance of the technology.

c. Challenges Identified:

The study also noted several challenges, such as facilitating conditions – having limited technological infrastructure and several inconsistencies that the current robots have. This have reflected negative impacts on the widespread adoption of robot kiosks in the Philippines. This means there needs to have better development before, during, and even after the deployment like consistent maintenance, having better infrastructure, and constant monitoring especially since these are yet widely established across the country.

d. Strategic Implications:

To ensure that robot kiosks are well-received and effective in retail environments, companies must tackle several important managerial challenges, including manufacturability, social, ethical, and economic factors which are typically constraints. Manufacturability as a constraint is a key challenge in deploying robot kiosks. Supporting infrastructure development is crucial for facilitating the deployment of robot kiosks. This includes improving digital connectivity and technological resources to ensure that kiosks can be effectively implemented across diverse regions. Additionally, establishing educational initiatives and training programs focused on the benefits and practical use of robot kiosks can enhance understanding and acceptance among businesses and consumers – leading to positive and higher satisfaction and adoption.

Addressing these challenges and fostering investment in AI and robotics are essential for further advancing these technologies, especially in developing smart systems and manufacturing. Overall, the findings of

this study aligned with the United Nations Sustainable Development Goal 9, which emphasizes industry innovation and infrastructure. Ongoing research, development, and strategic deployment of robot kiosks and service robots can drive technological progress, improve service quality, and contribute to socioeconomic development in the Philippines. As the country continues to embrace automation and sustainability, these insights will be crucial for optimizing the use of robot kiosks and meeting the evolving needs of consumers.

CRedit authorship contribution statement

Dioseph Andre F. Anduyo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ardvin Kester S. Ong:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Maela Madel L. Cahigas:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Ma. Janice J. Gumasing:** Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dhonn Q. Tomas:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.rineng.2025.105029](https://doi.org/10.1016/j.rineng.2025.105029).

Data availability

Data will be made available on request.

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